

Mahesh M

Data Engineer

mahesh709543@gmail.com | 959-888-8602

SUMMARY

Data Engineer with 4+ years of experience building scalable batch and streaming pipelines across healthcare, finance, and retail environments. Proficient in Azure Data Factory, Databricks, and Synapse for large-scale data processing over 15 terabytes, with expertise in Python, SQL, and Snowflake. Improved pipeline efficiency by 35%, reduced reporting latency by 50%, and increased forecasting accuracy by 15% through Power BI and Tableau. Skilled in data modeling, transforming structured and semi-structured data, and delivering compliant solutions for HIPAA, GDPR, AML, and KYC using Terraform, CI/CD, and Azure DevOps.

PROFESSIONAL EXPERIENCE

Data Engineer

Elevance Health | Texas

Aug 2024 – Present

- Developed ETL pipelines in Azure Data Factory and Databricks to process 5M+ healthcare records daily, improving efficiency by 40% through structured scheduling and error handling.
- Optimized SQL transformations in Snowflake and BigQuery, improving query speed by 30% and supporting faster access to clinical reporting data.
- Orchestrated end-to-end healthcare workflows with Apache Airflow, automating ETL pipelines for claims and patient encounter data, cutting manual intervention by 30%.
- Designed and deployed TensorFlow models to predict high-risk patients with 90% accuracy, reducing readmissions by 15% using Docker for packaging and Kubernetes for scalable deployment.
- Built HIPAA- and GDPR-compliant storage solutions using Azure Data Lake Gen2 and Blob Storage, managing 5+ TB of sensitive data and reducing retrieval time by 30%.
- Automated Azure infrastructure provisioning with Terraform, reducing environment setup time by 50% and ensuring consistency across development, testing, and production cloud environments.
- Developed interactive Power BI dashboards for clinical and operational analytics, delivering actionable insights to healthcare stakeholders and accelerating decision-making by 50%

Data Engineer

Barclays | New Jersey

Jun 2023 – Aug 2024

- Built streaming pipelines using Apache Kafka and AWS Lambda to process transactional events for fraud detection and AML risk scoring, reducing fraud by 25% and improving model accuracy by 30%.
- Optimized distributed data pipelines in Databricks using PySpark over 15TB of financial data, integrating Snowflake and BigQuery for high-performance querying, cutting processing time by 35%.
- Built a unified S3 data lake with AWS Glue Catalog to consolidate structured and unstructured data, cutting storage costs by 20% and making datasets easier to query for analytics teams.
- Automated data workflows using AWS Step Functions, Lambda, and Aurora (RDS), reducing manual processes by 30% and improving SQL query performance by 15% for high-volume payment and account data.
- Implemented access control and encryption using AWS IAM and KMS to meet SOX and PCI requirements, automated infrastructure with Terraform, integrated CI/CD workflows, and deployed Kubernetes workloads to improve deployment speed and reliability by 50%.

Data Analytics Engineer

Flipkart | Hyderabad, India

Sep 2020 – Dec 2022

- Built recommendation models using TensorFlow on Google AI Platform to personalize product suggestions, improving engagement by 18% and increasing conversions by 12% on the website and mobile app.
- Designed and optimized streaming data pipelines with GCP Dataflow, Kafka, and Databricks to process 2.5TB daily retail data, reducing latency in reporting workflows by 40%.
- Developed scalable ETL workflows using Apache Airflow and PySpark, reducing processing time by 30% and accelerating insight delivery for inventory and sales intelligence.
- Optimized big data storage and query performance in Google BigQuery, improving complex query response times by 40% and accelerating pricing and trend analysis for retail teams.
- Leveraged GCP services like Cloud Storage, Composer, and Dataflow to implement cost-efficient data architectures, reducing infrastructure spend by 30% while maintaining performance at scale.
- Created and maintained interactive Tableau dashboards for sales forecasting and inventory analytics, improving

business visibility and increasing forecast accuracy by 15%.

TECHNICAL SKILLS

Programming Languages: Python, SQL, T-SQL, PL/SQL, Shell Scripting (UNIX/Linux)

Data Processing: Apache Spark, Kafka, Hadoop (HDFS, MapReduce, Hive), Spark SQL, EDA, TensorFlow

ETL Tools: Apache Airflow, Talend, Informatica, Azure Data Factory

Cloud Platforms: AWS (S3, Glue, EMR, Lambda, Step Functions, CloudWatch), Azure (Data Factory, SQL Database, Data Lake), GCP (BigQuery, Dataflow, AI Platform)

Data Warehousing: Snowflake, Amazon Redshift, Google BigQuery, SQL Server, Hive

Database Management: MongoDB, PostgreSQL, MySQL

Data Visualization: Power BI, Tableau, Excel

CI/CD & DevOps: Jenkins, Docker, Kubernetes, Terraform, Azure DevOps

Data Governance: IAM Roles, RBAC, Dynamic Data Masking, Tag-Based Access Policies, HIPAA, AML, KYC, GDPR, CCPA

Tools & Collaboration: JIRA, Confluence, GitHub, Visual Studio Code, Agile Methodologies

EDUCATION

University of New Haven | CT

Hindustan Institute of Technology | Coimbatore, India